

DEPARTMENT OF MATHEMATICS
MAT111 INTRODUCTORY MATHEMATICS I

WEEK 4: LECTURE NOTE

REVISION OF THE CARTESIAN PLANE

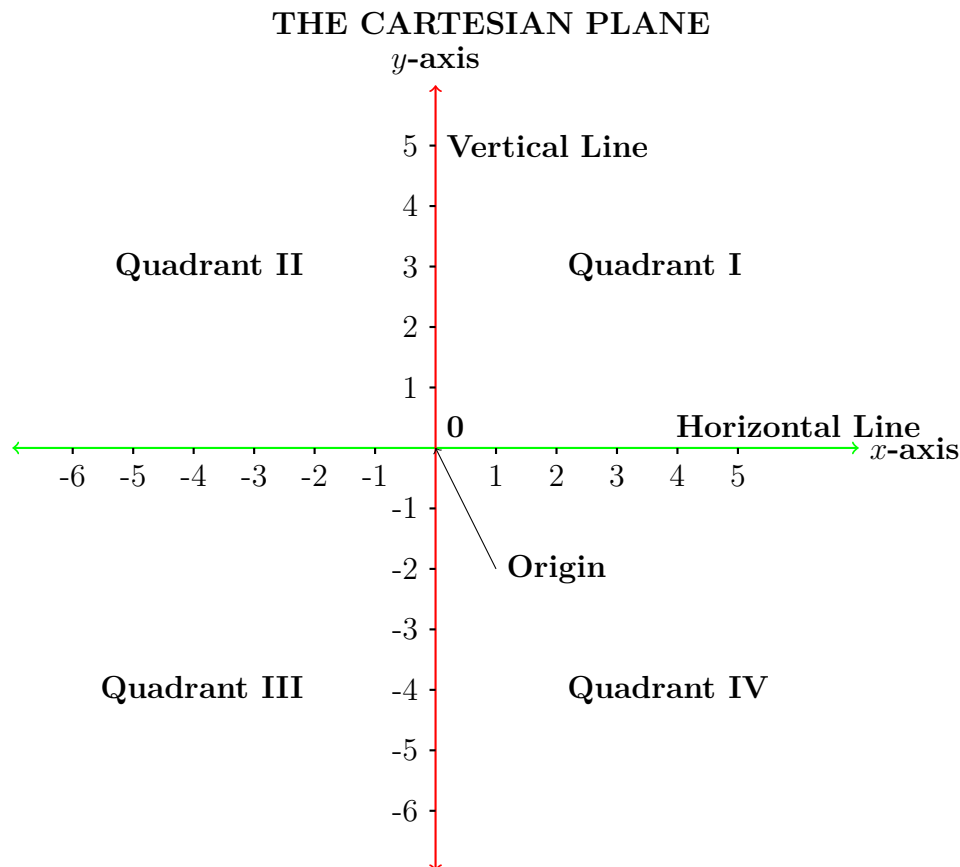
Just as we can represent real numbers by points on a real number line, we can represent **ordered pairs of real numbers** by points in a plane called **the rectangular coordinate system, or the Cartesian plane**, after the French mathematician René Descartes (1596-1650).

The cartesian plane is formed by using two real number lines intersecting at right angles. The horizontal line is usually called the **x -axis**, and the vertical line is usually called the **y -axis**. The point of intersection of these two axis is the **origin**(0, 0), and the two axes divide the plane into four parts called **quadrants**.

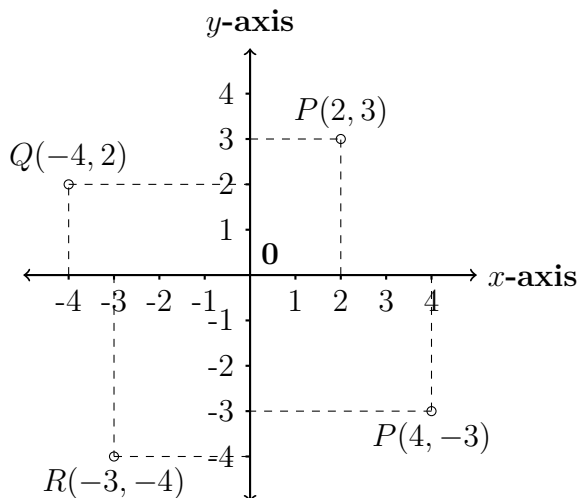
Each point in the plane corresponds to an ordered pair (x, y) of real numbers x and y , called **coordinates of the point**. The x -coordinate represents the directed distance from the y -axis to the point, and the y -coordinate represents the directed distance from the x -axis to the point.

Directed distance from y -axis Directed distance from x -axis

(x, y)



PLOTTING POINTS ON THE CARTESIAN PLANE



Using the Pythagorean Theorem for a right triangle we can easily formulate the following theorem.

Theorem 1 (Distance Formula). *The distance d between the points $P(x_1, y_1)$ and $Q(x_2, y_2)$ in the plane is*

$$PQ = d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Example 2 (Finding distance). *Find the distance between the points $P(-2, 1)$ and $Q(3, 4)$ on the plane.*

Solution. *Note that here $x_1 = -2$, $y_1 = 1$, $x_2 = 3$ and $y_2 = 4$. So that*

$$\begin{aligned} PQ = d &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{[3 - (-2)]^2 + (4 - 1)^2} = \sqrt{34} \approx 5.83 \end{aligned}$$

Example 3 (Verifying a Right Triangle). *Show that the points $P(2, 1)$, $Q(4, 0)$ and $R(5, 7)$ are the vertices of a right triangle.*

Solution. *Lengths of sides of $\triangle PQR$ are given by*

$$\begin{aligned} d_1 = PQ &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{[4 - 2]^2 + (0 - 1)^2} = \sqrt{5} \\ d_2 = PR &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{[5 - 2]^2 + (7 - 1)^2} = \sqrt{45} \\ d_3 = QR &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{[5 - 4]^2 + (7 - 0)^2} = \sqrt{50} \end{aligned}$$

Since $d_3 > d_1$, $d_2 > d_1$ and $d_3 > d_2$, PQR is a triangle. Because

$$d_1^2 + d_2^2 = 5 + 45 = 50 = d_3^2$$

by the Pythagorean Theorem we conclude that $\triangle PQR$ is a right angled triangle with hypotenuse QR and right angle $\angle QPR$.

Theorem 4. *The midpoint of the line segment joining the points (x_1, y_1) and (x_2, y_2) is given by the Midpoint Formula:*

$$\text{Midpoint} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$

Example 5 (Finding a Line Segment's Midpoint). Find the midpoint of the line segment joining the points $P(-5, -3)$ and $Q(9, 3)$.

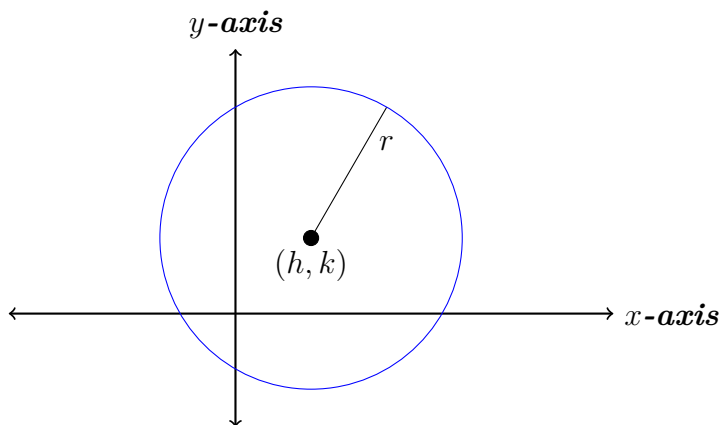
Solution. Let $(x_1, y_1) = (-5, -3)$ and $(x_2, y_2) = (9, 3)$.

$$\begin{aligned}\text{Midpoint} &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) && \text{Midpoint Formula} \\ &= \left(\frac{-5 + 9}{2}, \frac{-3 + 3}{2} \right) = (2, 0).\end{aligned}$$

Theorem 6. The **standard form of the equation of a circle** is

$$(x - h)^2 + (y - k)^2 = r^2.$$

The point (h, k) is the **center of the circle**, and the positive number r is the **radius** of the circle. The standard form of the equation of a circle whose center is at the origin, is $x^2 + y^2 = r^2$.



Example 7 (Writing an Equation of a Circle). The point $P(3, 4)$ lies on a circle whose center is at $C(-1, 2)$. Write the standard form of the equation of this circle.

Solution. The radius r of the circle is the distance between $C(-1, 2)$ and $P(3, 4)$. Therefore,

$$\begin{aligned}r &= \sqrt{[3 - (-1)]^2 + (4 - 2)^2} && \text{Distance Formula} \\ &= \sqrt{4^2 + 2^2} && \text{Simplify} \\ &= \sqrt{20} && \text{Radius}\end{aligned}$$

The equation of the circle with center $(h, k) = (-1, 2)$ and $r = \sqrt{20}$ is

$$\begin{aligned}(x - h)^2 + (y - k)^2 &= r^2 && \text{Equation of circle} \\ [x - (-1)]^2 + (y - 2)^2 &= (\sqrt{20})^2 && \text{Substitute for } h, k, r \\ (x + 1)^2 + (y - 2)^2 &= 20 && \text{Standard form}\end{aligned}$$

Example 8 (Writing an Equation of a Circle). Write the standard form of the equation of the circle with center $(2, -1)$ radius 5 units.

Solution. Note that here $(h, k) = (2, -1)$ and $r = 5$, so that

$$\begin{aligned}(x - h)^2 + (y - k)^2 &= r^2 && \text{Equation of circle} \\ [x - 2]^2 + [y - (-1)]^2 &= 5^2 && \text{Substitute for } h, k, r \\ (x - 2)^2 + (y + 1)^2 &= 25 && \text{Standard form}\end{aligned}$$

Example 9 (Writing an Equation of a Circle). Write the standard form of the equation of the circle with endpoints of a diameter are $A(-4, 2)$ and $B(3, 5)$.

Solution. Note that diameter of a circle is twice the radius and the midpoint of a diameter is the center of the circle.

Thus we have

$$(h, k) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) = \left(\frac{-4 + 3}{2}, \frac{2 + 5}{2} \right) = \left(\frac{-1}{2}, \frac{7}{2} \right)$$

$$r = \frac{1}{2}AB = \frac{1}{2}\sqrt{[3 - (-4)]^2 + [5 - 2]^2} = \frac{\sqrt{58}}{2}$$

Hence the standard form of equation of the circle is given by

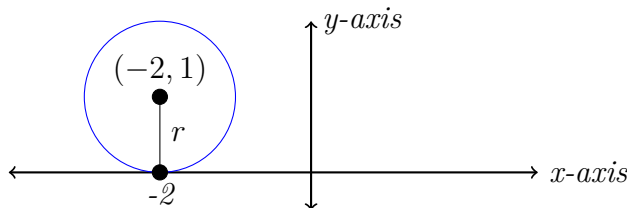
$$(x - h)^2 + (y - k)^2 = r^2 \quad \text{Equation of circle}$$

$$\left[x - \left(\frac{-1}{2} \right) \right]^2 + \left[y - \frac{7}{2} \right]^2 = \left(\frac{\sqrt{58}}{2} \right)^2 \quad \text{Substitute for } h, k, r$$

$$\left(x + \frac{1}{2} \right)^2 + \left(y - \frac{7}{2} \right)^2 = \frac{58}{4} = \frac{29}{2} \quad \text{Standard form}$$

Example 10. Write the equation of a circle with center $(-2, 1)$ and tangent to the x -axis.

Solution. Since the x -axis is tangent to the circle, the distance from x -axis to the center is the radius of the circle.



Since $(h, k) = (-2, 1)$ and $r = 1$, we have

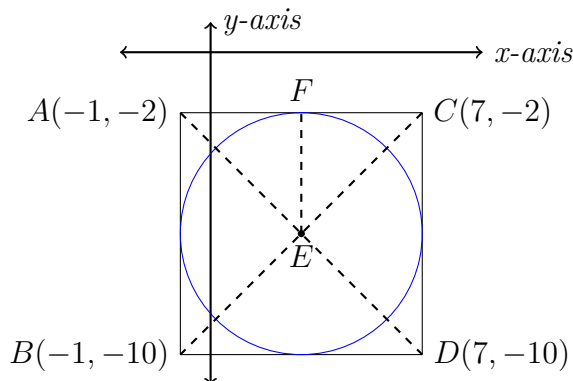
$$(x - h)^2 + (y - k)^2 = r^2 \quad \text{Equation of circle}$$

$$[x - (-2)]^2 + [y - 1]^2 = 1^2 \quad \text{Substitute for } h, k, r$$

$$(x + 2)^2 + (y - 1)^2 = 1 \quad \text{Standard form}$$

Example 11. Write the standard equation of a circle inscribed in the square with vertices $A(-1, -2)$, $B(-1, -10)$, $C(7, -2)$ and $D(7, -10)$.

Solution. The sides of the square are tangent to the circle.



The Midpoint of the points A and D is the center of the circle. Thus

$$(h, k) = \left(\frac{-1+7}{2}, \frac{-2-10}{2} \right) = (3, -6).$$

The distance from Midpoint of $\overline{AC}$ to E is the radius of the circle.

$$\text{Midpoint of } \overline{AC} = F = \left(\frac{-1+7}{2}, \frac{-2-2}{2} \right) = (3, -2).$$

Thus we have,

$$r = \sqrt{(3-3)^2 + (-6-(-2))^2} = \sqrt{0+16} = \sqrt{16} = 4.$$

Therefore, the standard form of the equation of the circle is

$(x-h)^2 + (y-k)^2 = r^2$	Equation of circle
$[x-3]^2 + [y-(-6)]^2 = 4^2$	Substitute for h, k, r
$(x-3)^2 + (y+6)^2 = 16$	Standard form

LINEAR FUNCTIONS

A graph of a function of the form

$$y = f(x) = mx + b,$$

where m and b are real numbers, is a straight line whose slope is m and whose y -intercept is $(0, b)$.

Definition 12 (The Slope of a Line). *The **slope** m of the nonvertical line through (x_1, y_1) and (x_2, y_2) is*

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{change in } y}{\text{change in } x}, \text{ where } x_1 \neq x_2.$$

Example 13 (Finding the Slope of a Line). *Find the slope of the line passing through $(-2, 0)$ and $(3, 1)$.*

Solution. $m = \frac{1-0}{3-(-2)} = \frac{1}{5}.$

Example 14 (Finding the Slope of a Line). *Find the slope of the line passing through $(-1, 2)$ and $(2, 2)$.*

Solution. $m = \frac{2-2}{2-(-1)} = \frac{0}{3} = 0.$

Example 15 (Finding the Slope of a Line). *Find the slope of the line passing through $(0, 4)$ and $(1, -1)$.*

Solution. $m = \frac{-1-4}{1-0} = \frac{-5}{1} = -5.$

The Slope of a Line

1. A line with positive slope ($m > 0$) rises from left to right.

2. A line with negative slope ($m < 0$) falls from left to right.
3. A line with zero ($m = 0$) slope is horizontal.
4. A line with undefined slope is vertical.

Definition 16 (Point-Slope Form of the Equation of a Line). *The point-slope form of the equation of the line that passes through the point (x_1, y_1) and has a slope of m is*

$$y - y_1 = m(x - x_1).$$

Example 17 (The Point-Slope Form of the Equation of a Line). *Find an equation of the line that passes through the point $(1, -2)$ and has a slope of 3.*

Solution.

$$y - y_1 = m(x - x_1)$$

$$y - (-2) = 3(x - 1)$$

$$y + 2 = 3x - 3$$

$$y = 3x - 5$$

Definition 18 (Parallel Lines). *Two distinct non-vertical lines ℓ_1 and ℓ_2 with slopes m_1 and m_2 , respectively, are **parallel**, denoted by $\ell_1 \parallel \ell_2$, if and only if their slopes are equal, that is,*

$$m_1 = m_2.$$

Example 19 (Equations of Parallel Lines). *Find the equation of the line that passes through the point $(2, 1)$ and parallel to the line $2x - 3y = 5$.*

Solution. *Begin by writing the equation of the line in the form $y = mx + c$.*

$$2x - 3y = 5$$

$$-3y = -2x + 5$$

$$y = -\frac{2}{-3}x + \frac{5}{-3}$$

$$\boxed{y = \frac{2}{3}x - \frac{5}{3}}$$

Therefore the given line has a slope of $m_1 = \frac{2}{3}$. Any line parallel to the given line must also have a slope of $m_2 = \frac{2}{3}$. So, the line through $(2, -1)$ has the following equation

$$y - y_1 = m_2(x - x_1)$$

$$y - (-1) = \frac{2}{3}(x - 2)$$

$$y + 1 = \frac{2}{3}x - \frac{4}{3}$$

$$y = \frac{2}{3}x - \frac{7}{3}.$$

Definition 20 (Perpendicular Lines). *Two distinct non-vertical lines ℓ_1 and ℓ_2 with slopes m_1 and m_2 , respectively, are **perpendicular**, denoted by $\ell_1 \perp \ell_2$, if and only if*

$$m_1 m_2 = -1, \quad \text{or} \quad m_2 = -\frac{1}{m_1}.$$

Example 21 (Equations of Perpendicular Lines). *Find the equation of the line that passes through the point $(2, -1)$ and is perpendicular to the line $2x - 3y = 5$.*

Solution. *From the preceding example, you know that the given equation can be written as $y = \frac{2}{3}x - \frac{5}{3}$. So, any line perpendicular to this line must have a slope of*

$$m_2 = -\frac{1}{m_1} = -\frac{3}{2}.$$

So, the line through the point $(2, -1)$ is

$$y - (-1) = -\frac{3}{2}(x - 2)$$

$$y + 1 = -\frac{3}{2}x + 3$$

$$y = -\frac{3}{2}x + 2$$

SUMMARY OF EQUATIONS OF LINES.

1. General form of the equation of a line

$$Ax + By + C = 0,$$

where A and B are not both zero.

2. Vertical line

$$x = a$$

3. Horizontal line

$$y = b$$

4. Slope-intercept form

$$y = mx + b$$

5. Point-slope form

$$y - y_1 = m(x - x_1)$$